Analyzing Traffic Accident Patterns in Southern Interior BC

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Research Questions

1. Which municipalities in the Southern Interior have the highest crash rates per capita?

2. Do specific accident types show significant differences during certain months or times of day? Are there seasonal patterns for accident types?

3. Are there specific high-risk zones for vulnerable users based on intersection density and time of day?

4. Are accidents clustered at any intersections, roads, or communities?

Pipeline Section 1: Data Acquisition and Loading

Crashes

ICBC Reported Crashes

- Accidents reported to ICBC
- Open Data License
- 2020-2024
- 200,981 records
  - total of 202,472 accidents
- 25 Columns
  - Temporal
    - Year, month, day, time
  - Descriptive
    - Animal, pedestrian, cyclist, heavy vehicle, severity
  - Spatial
    - Latitude, longitude
    - Street name, municipality name
  - Numeric
    - Total crashes
    - Total victims

Pipeline Section 1: Data Acquisition and Loading

Population Estimates & Projections

- BC Stats
- Open Government License
- Estimated populations per year
- Projected future populations
- 191 records
- 5 Columns
  - Region
  - Municipality
  - Year
  - Type
  - Gender
    - We used total
  - Total

Section 2: ETL, Data Cleaning

Populations

- Minimal cleaning
- Missing values
  - Latitude/Longitide
  - Street names
  - Cross street names
- Load
  - Import into SQLite table using Python
  - Converted numeric values

	column_name	column_dtype	num_nulls	num_non_nulls	num_of_distinct_values	num_empty_strings
0	id	int64	0	200981	200981	0
1	date_of_loss_year	int64	0	200981	5	0
2	crash_severity	str	0	200981	2	0
3	cyclist_flag	str	0	200981	2	0
4	day_of_week	str	0	200981	7	0
5	derived_crash_config	str	0	200981	11	0
6	heavy_veh_flag	str	0	200981	2	0
7	intersection_crash	str	0	200981	2	0
8	month_of_year	str	0	200981	12	0
9	motorcycle_flag	str	0	200981	2	0
10	parked_vehicle_flag	str	0	200981	2	0
11	parking_lot_flag	str	0	200981	2	0
12	pedestrian_flag	str	0	200981	2	0
13	region	str	0	200981	1	0
14	time_category	str	0	200981	8	0
15	municipality	str	0	200981	321	0
16	road_location_description	str	0	200981	24777	0
17	street_full_name	str	0	200981	14140	43
18	total_crashes	int64	0	200981	5	0
19	total_victims	int64	0	200981	13	0
20	animal_flag	str	0	200981	2	0
21	cross_street_full_name	str	0	200981	7804	138149
22	latitude	float64	70850	130131	24222	0
23	longitude	float64	70850	130131	24408	0
24	mid_block_crash	str	0	200981	2	0

Section 2: ETL, Data Cleaning

Populations

- Cleaning
  - Skipped metadata rows
  - Removed thousand separator
- Load
  - Import into SQLite table using Python
- Filter
  - Only kept municipalities present in Reported Crashes

Population	55.0	11847.0	27087.0	209.0	1761.0	4289.0	8706.0	165210.0

	column_name	column_dtype	num_nulls	num_non_nulls	num_of_distinct_values	num_empty_strings
0	region	str	0	55	55	0
1	municipality	str	0	55	55	0
2	total	int64	0	55	55	0

Section 2: ETL, Data Cleaning

Crashes-Join-Populations

- Joined crashes to population in new SQLite table on municipality
- Calculated sums of crashes and of victims
- Calculated rate of crashes and victims compared to population
  - crash rate = crashes / population * 100000
  - victims rate = victims / population * 100000

	column_name	column_dtype	num_nulls	num_non_nulls	num_of_distinct_values	num_empty_strings
0	municipality	str	0	55	55	0
1	total_crashes	int64	0	55	54	0
2	total_victims	int64	0	55	52	0
3	crashes_per_100k	float64	0	55	55	0
4	victims_per_100k	float64	0	55	55	0
5	population	int64	0	55	55	0

Section 3: Data Storage

SQLite and Python

- db.py - create, populate, export data
- app.py - controls database layers

Tables

Search tables...

crashes 200,981

crashes_join_population 55

population 55

sqlite_sequence 1

crashes

Refresh | Schema | Export CSV | Export JSON | Copy All

SQL Query Editor (click to expand)

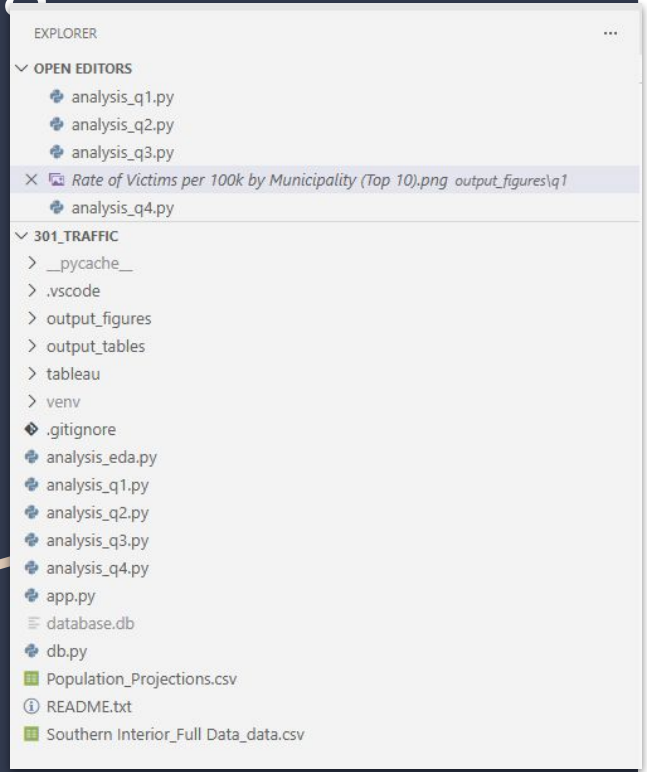
Filter rows...

200,981 total rows

ID	DATE_OF_LOSS_YEAR	CRASH_SEVERITY	CYCLIST_FLAG	DAY_OF_WEEK	DERIVED_CRASH_CONFIG	HEAVY_VEH_FLAG	INTERSECTION_CRASH
1	2024	CASUALTY CRASH	No	FRIDAY	SINGLE VEHICLE	No	No
2	2024	CASUALTY CRASH	No	FRIDAY	REAR END	No	No
3	2024	CASUALTY CRASH	No	FRIDAY	SIDE SWIPE - SAME DIRECTION	No	No
4	2024	CASUALTY CRASH	No	FRIDAY	SINGLE VEHICLE	No	No
5	2024	CASUALTY CRASH	No	FRIDAY	SINGLE VEHICLE	No	No
6	2024	CASUALTY CRASH	No	FRIDAY	SINGLE VEHICLE	No	No
7	2024	CASUALTY CRASH	No	FRIDAY	SINGLE VEHICLE	No	No

« ‹ Previous Page 1 of 2010 Next › » 100 rows

Section 3: Data Storage



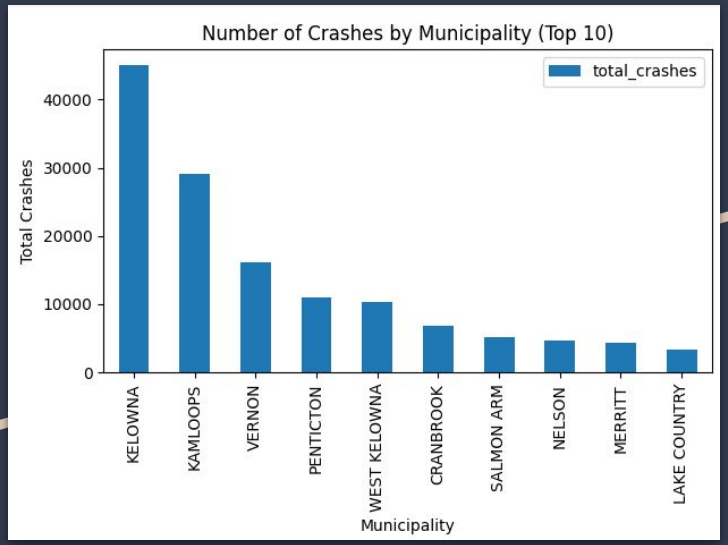
SQLite and Python

- db.py - create, populate, export data
- app.py - controls database layers

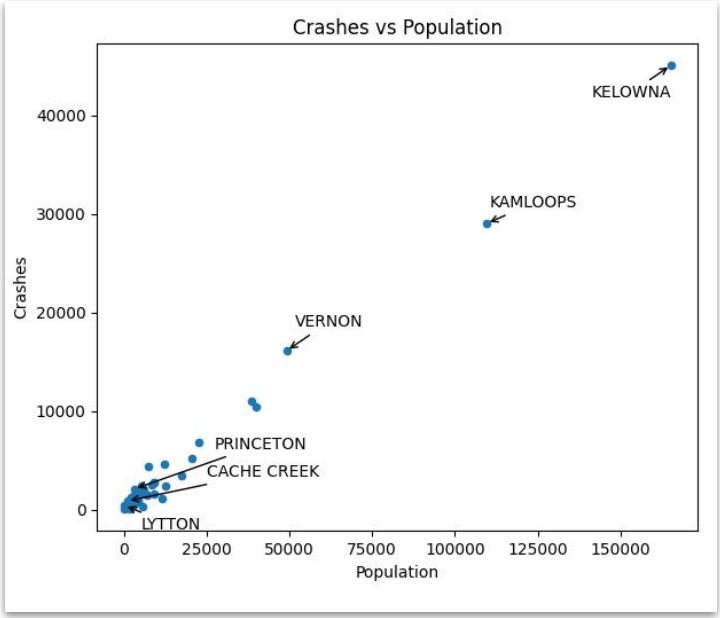
https://github.com/Badger-Tree/301_traffic

Section 4: Exploratory Data Analysis

Q1: Crash Rates and Victim Rates Per Municipality



- Higher population -> more accidents
- Rate of accidents and victims
  - crash rate = crashes / population * 100000
  - victims rate = victims / population * 100000



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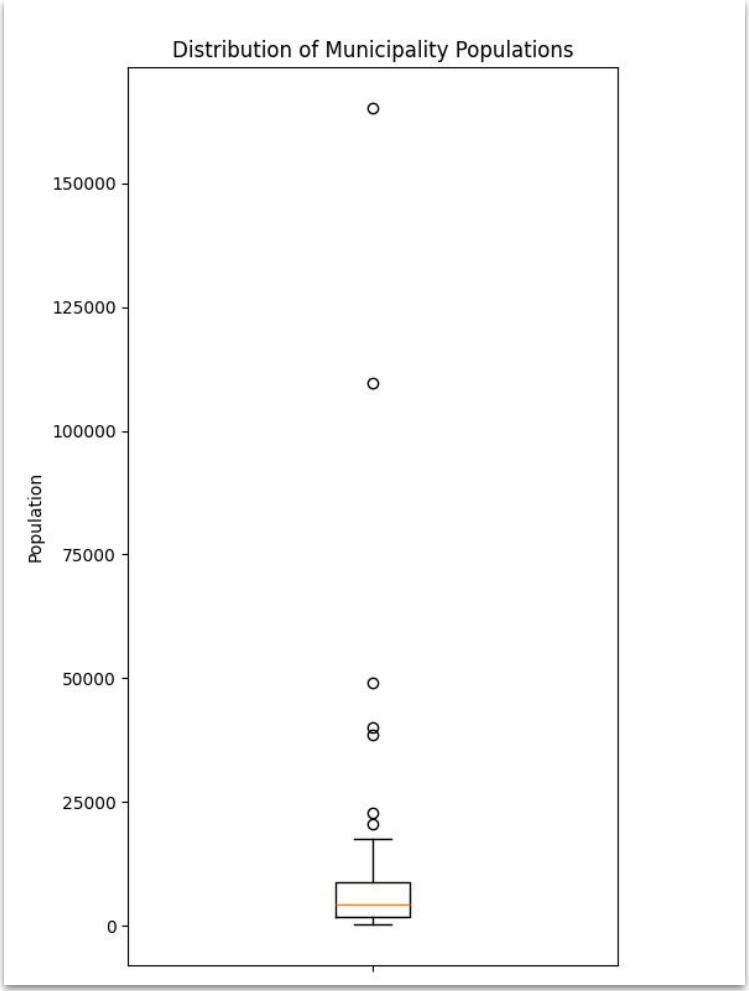
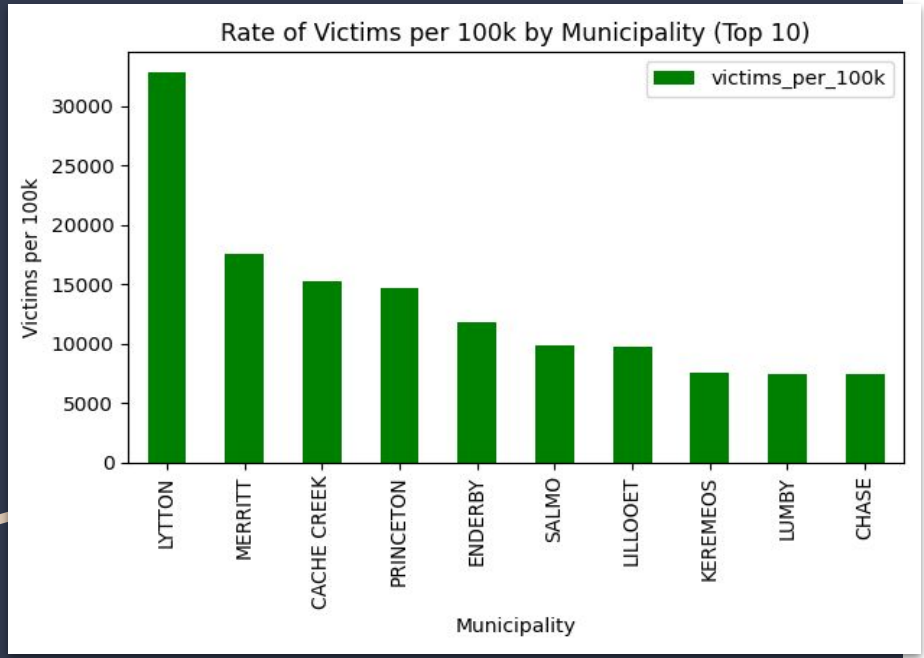
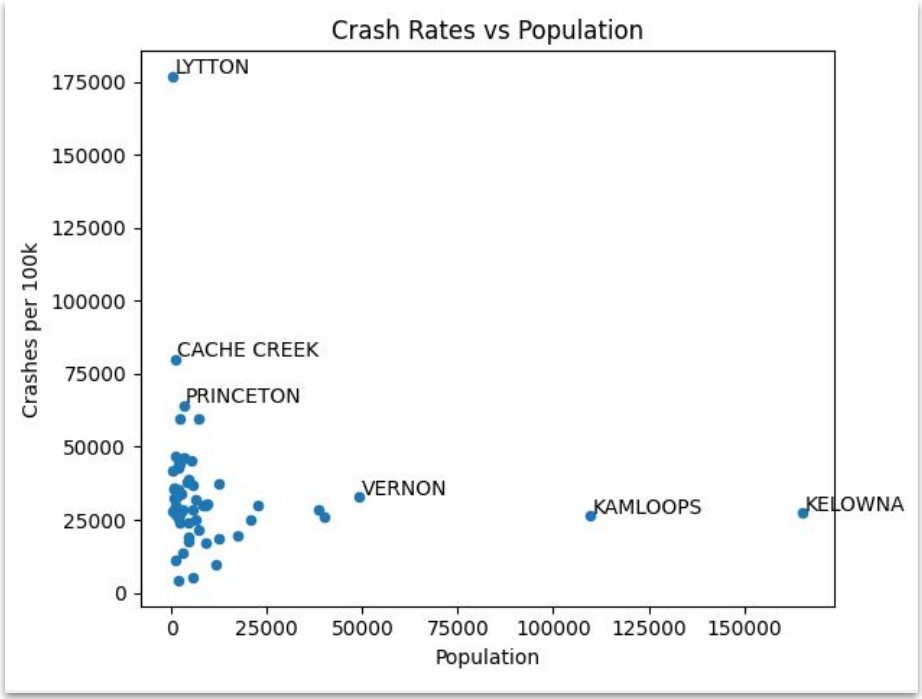


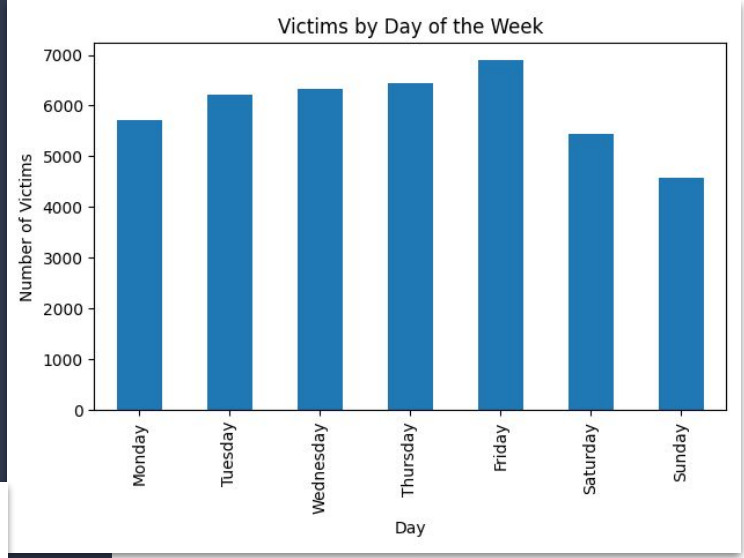
Q1: Crash Rates and Victim Rates Per Municipality

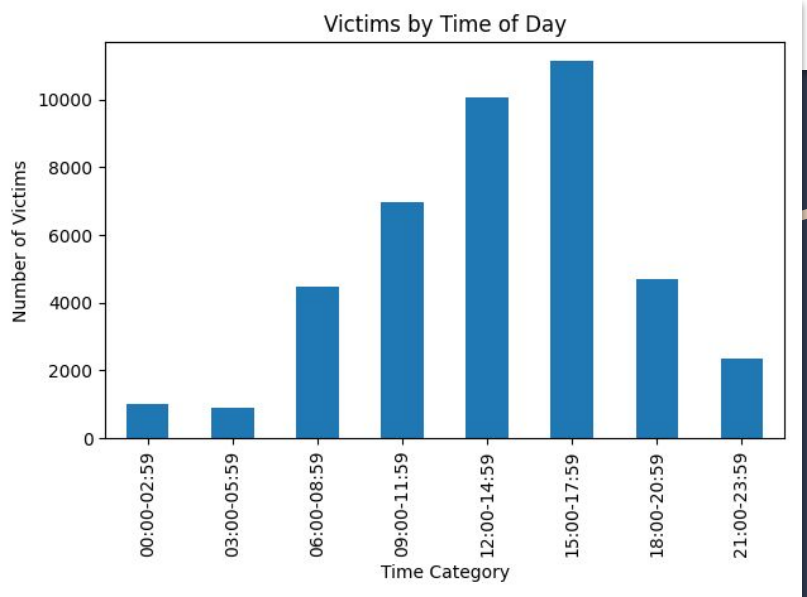
Highest rate -> Small towns





Q2: Influences of Time, Day, Month on Accidents





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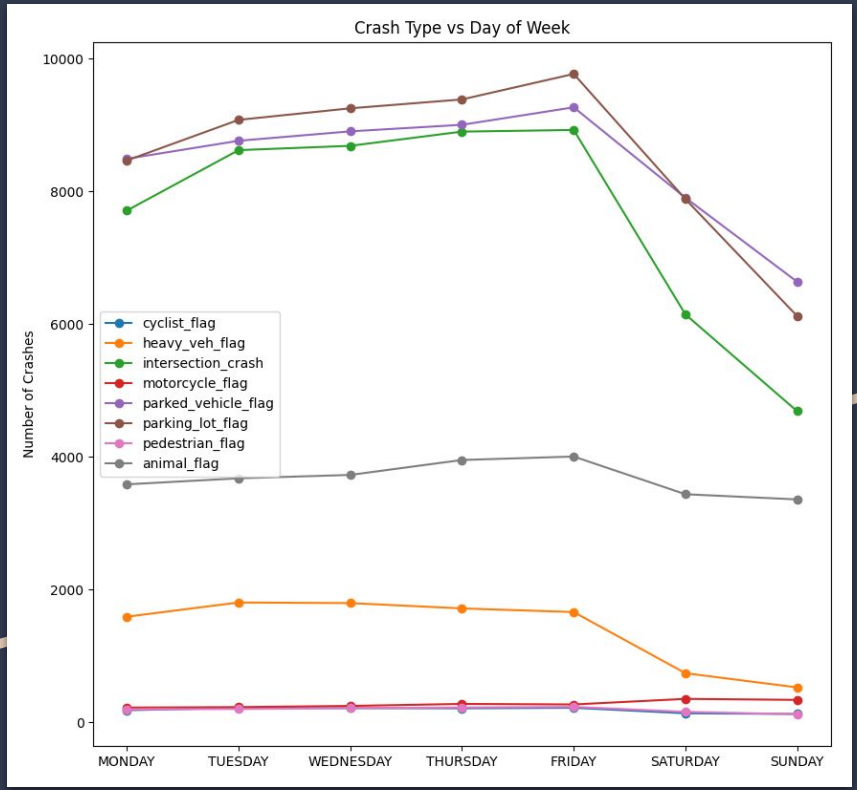
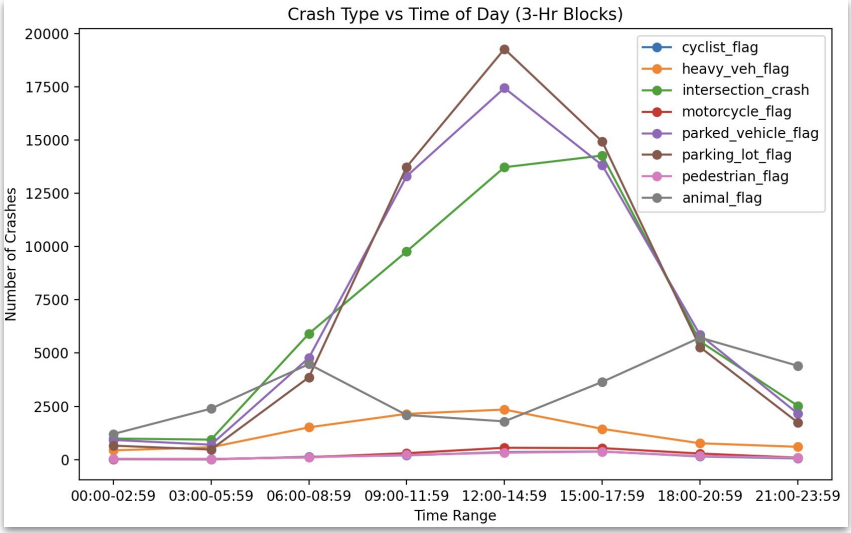
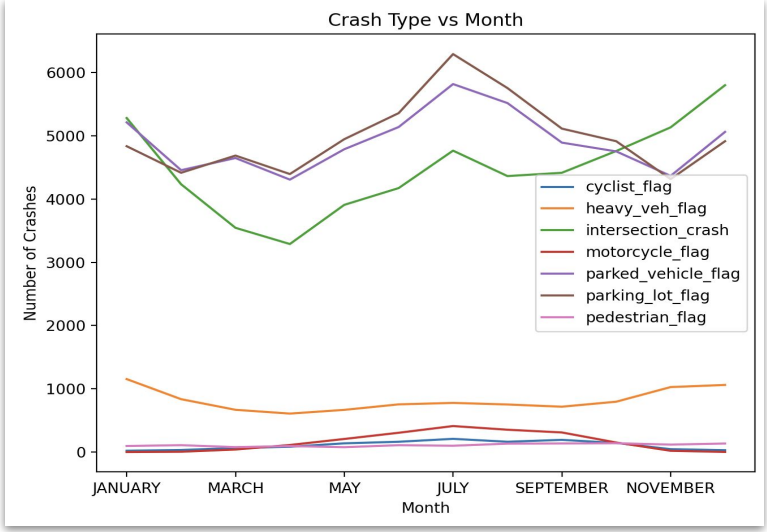
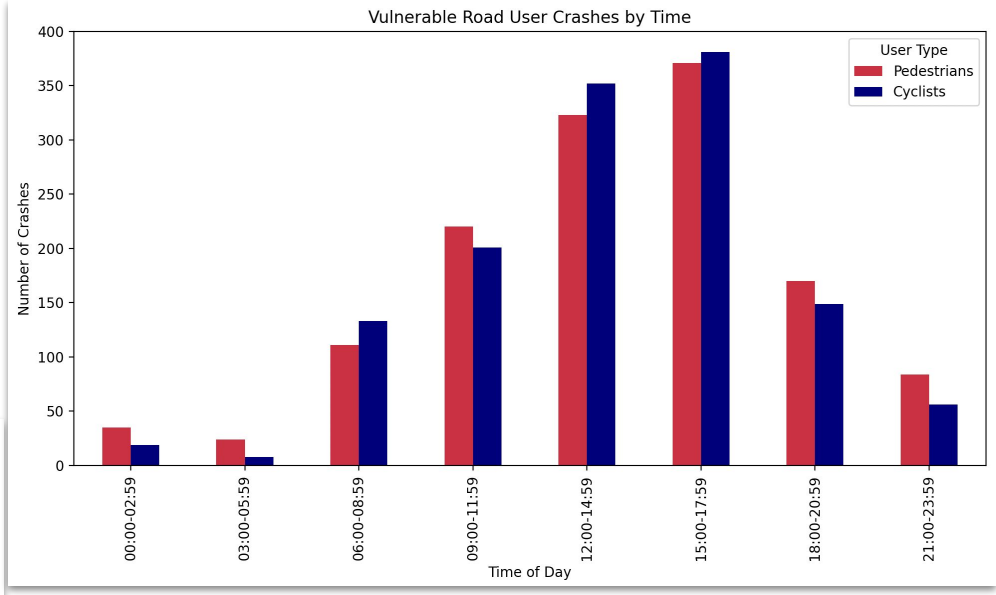
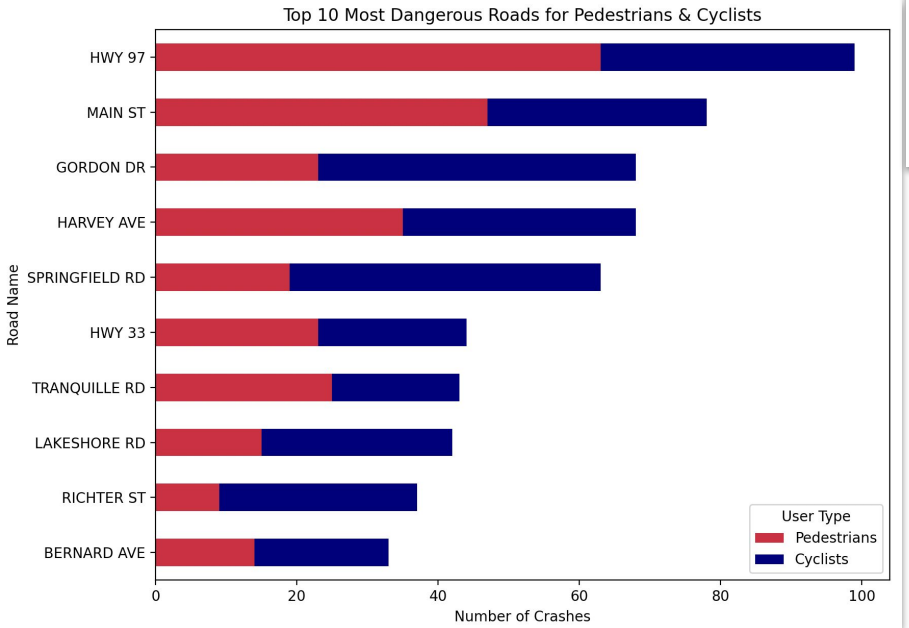






Q3: Risk Factors for Vulnerable Road Users





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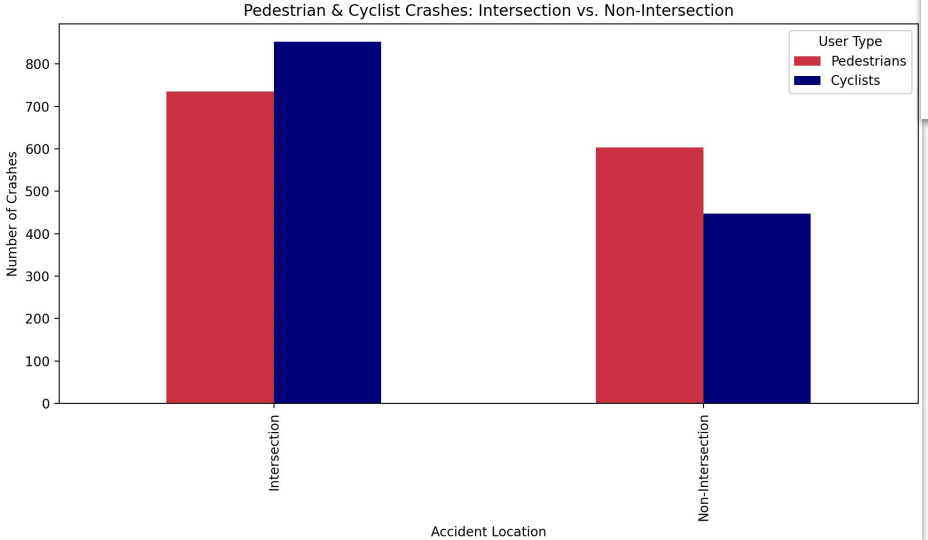
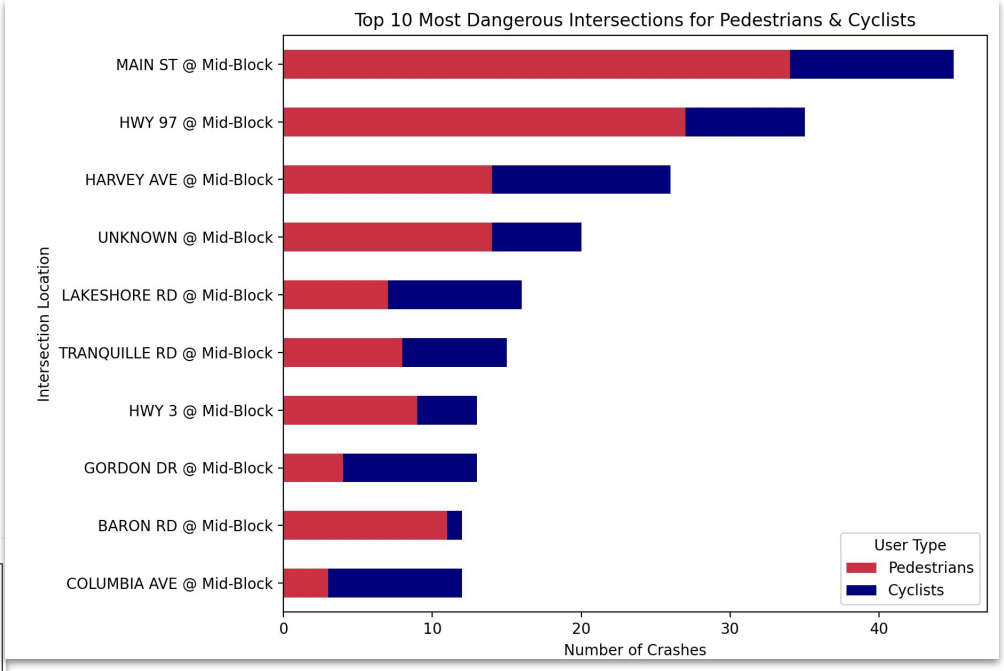
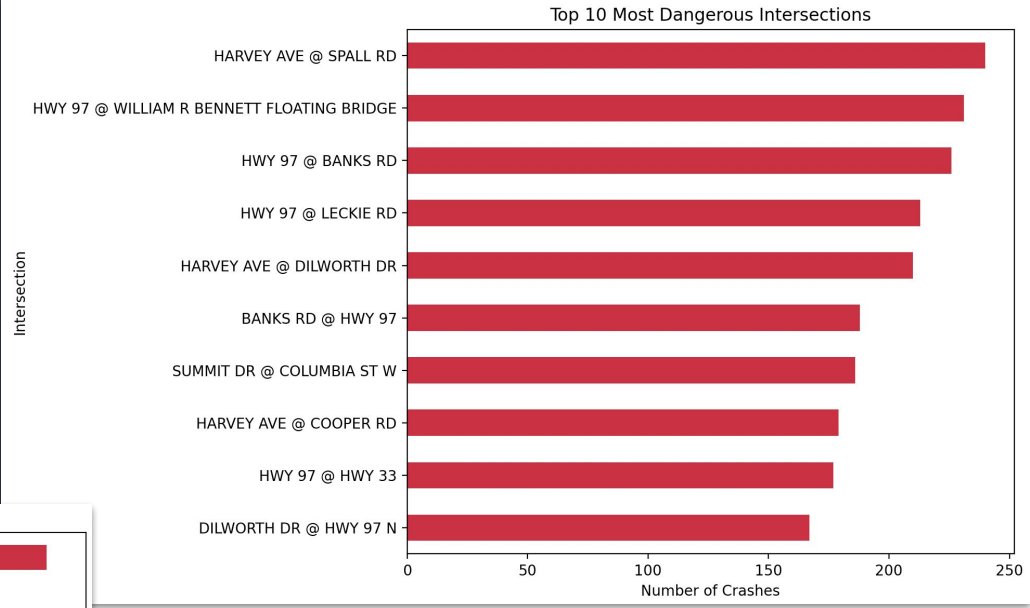
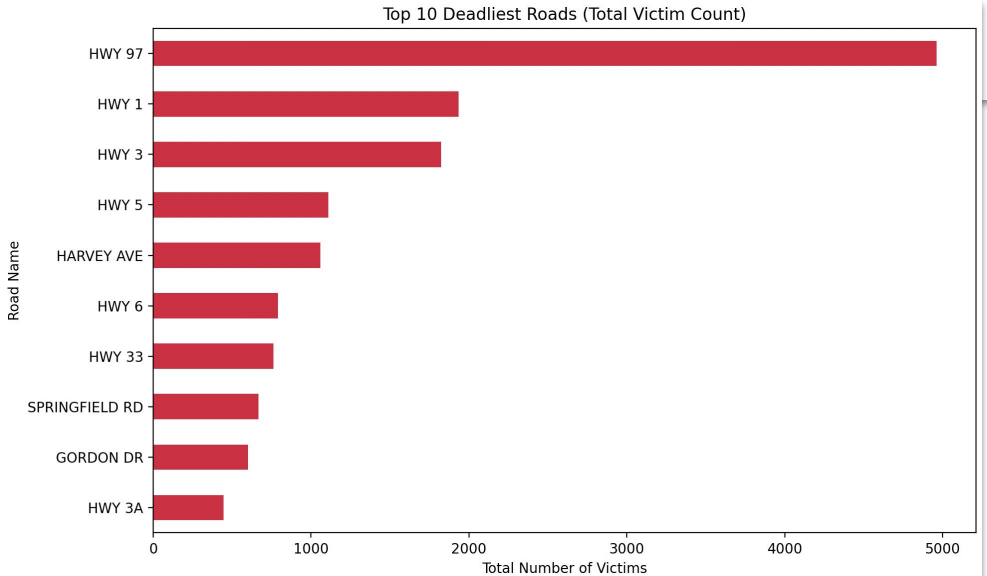




Q4: Spatial Hotspots of Accidents

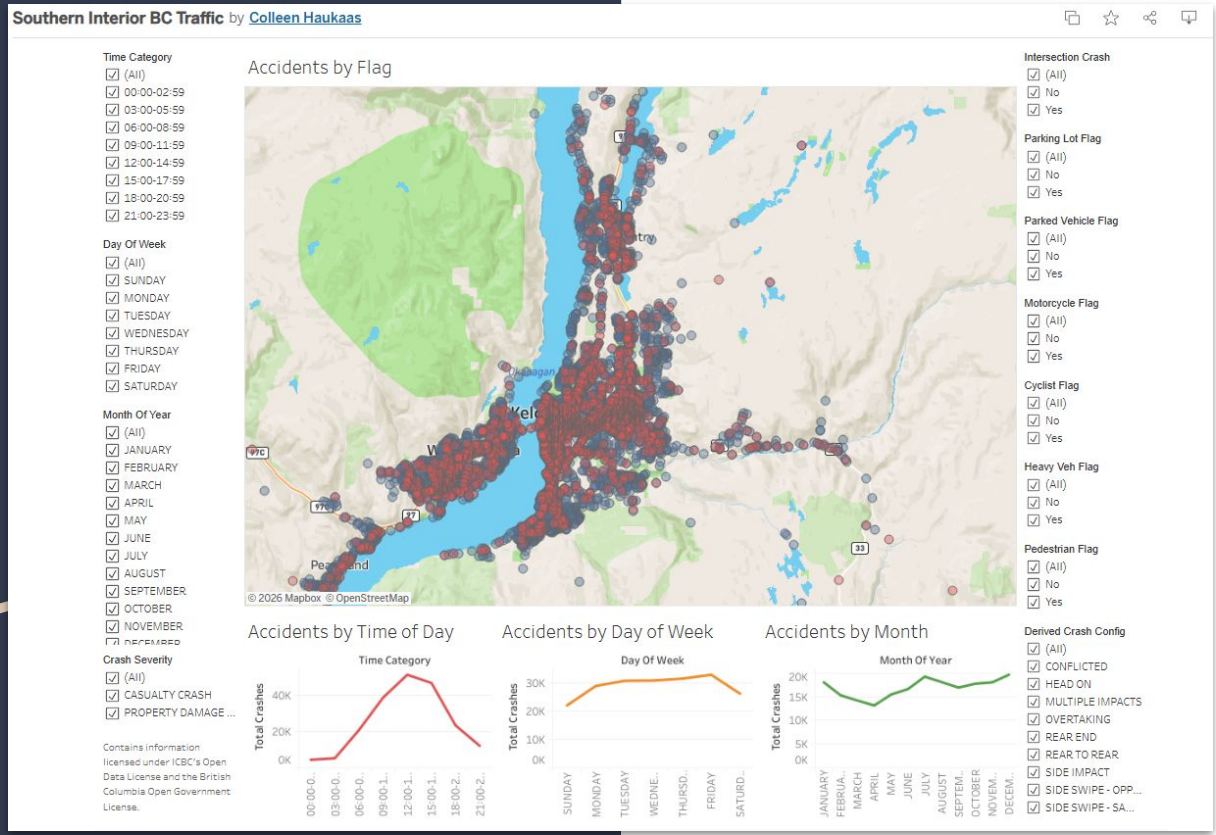




Visualization

Highlights on Tableau

https://public.tableau.com/app/profile/colleen.haukaas/viz/SouthernInteriorBCTraffic/Traffic_Dashboard



Conclusion and Next Steps

Analysis highlighted key collision factors:

- Larger municipalities (Kelowna, Kamloops) have highest volume of crashes
- Smaller towns like Lytton have higher rates of crashes;
  - proximity of highway junctions.
- Pedestrians and cyclists deal with different types of risks:
  - cyclists at intersections (possibly due to turning conflicts)
  - pedestrians in mid-block accidents (marked crosswalks but not at intersections).

Next steps:

- integrate traffic flow data, map mid-block hotspots, lighting and weather analysis to measure visibility on severity of crashes.